CLASS X : CHAPTER - 7
COORDINATE GEOMETRY

IMPORTANT FORMULAS & CONCEPTS

Points to remember

- The distance of a point from the *y*-axis is called its ***x*-coordinate**, or **abscissa**.
- The distance of a point from the *x*-axis is called its ***y*-coordinate**, or **ordinate**.
- The coordinates of a point on the *x*-axis are of the form $(x, 0)$.
- The coordinates of a point on the *y*-axis are of the form $(0, y)$.

Distance Formula

The distance between any two points $A(x_1, y_1)$ and $B(x_2, y_2)$ is given by

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$or \quad AB = \sqrt{(\text{difference of abscissae})^2 + (\text{difference of ordinates})^2}$$

Distance of a point from origin

The distance of a point P(x, y) from origin O is given by OP = $\sqrt{x^2 + y^2}$

Problems based on geometrical figure

To show that a given figure is a

- Parallelogram – prove that the opposite sides are equal
- Rectangle – prove that the opposite sides are equal and the diagonals are equal.
- Parallelogram but not rectangle – prove that the opposite sides are equal and the diagonals are not equal.
- Rhombus – prove that the four sides are equal
- Square – prove that the four sides are equal and the diagonals are equal.
- Rhombus but not square – prove that the four sides are equal and the diagonals are not equal.
- Isosceles triangle – prove any two sides are equal.
- Equilateral triangle – prove that all three sides are equal.
- Right triangle – prove that sides of triangle satisfies Pythagoras theorem.

Section formula

The coordinates of the point $P(x, y)$ which divides the line segment joining the points $A(x_1, y_1)$ and $B(x_2, y_2)$, internally, in the ratio $m_1 : m_2$ are

$$\left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$$

This is known as the **section formula**.

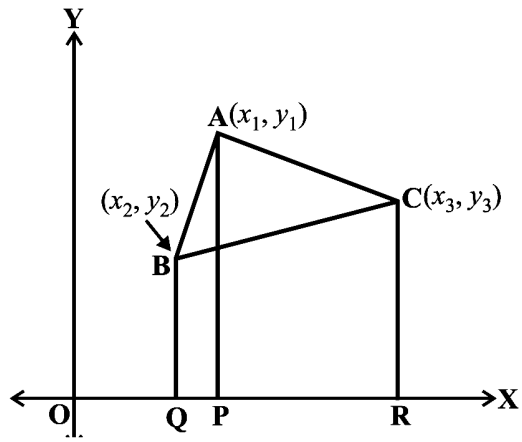
Mid-point formula

The coordinates of the point $P(x, y)$ which is the midpoint of the line segment joining the points $A(x_1, y_1)$ and $B(x_2, y_2)$, are $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

Area of a Triangle

If $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ are the vertices of a ΔABC, then the area of ΔABC is given by

$$Area\ \ of\ \ \Delta ABC = \frac{1}{2}[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]$$



Trick to remember the formula

The formula of area of a triangle can be learn with the help of following arrow diagram:

$$\Delta ABC = \frac{1}{2}\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \\ x_1 & y_1 \end{vmatrix}$$

Find the sum of products of numbers at the ends of the lines pointing downwards and then subtract the sum of products of numbers at the ends of the line pointing upwards, multiply the difference by $\frac{1}{2}$. .e. $Area\ \ of\ \ \Delta ABC = \frac{1}{2}[(x_1y_2 + x_2y_3 + x_3y_1) - (x_1y_3 + x_3y_2 + x_2y_1]$

MCQ WORKSHEET-I
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1. The points A(0, –2), B(3, 1), C(0, 4) and D(–3, 1) are the vertices of a
(a) parallelogram (b) rectangle (c) square (d) rhombus

2. If A(3, 8), B(4, –2) and C(5, –1) are the vertices of ΔABC. Then, its area is
(a) $28\frac{1}{2}$ sq. units (b) $37\frac{1}{2}$ sq. units (c) 57 sq. units (d) 75 sq. units

3. The points A(0, 6), B(–5, 3) and C(3, 1) are the vertices of a triangle which is
(a) isosceles (b) equilateral (c) scalene (d) right angled

4. Two vertices of ΔABC are A(–1, 4) and B(5, 2) and its centroid is G(0, –3). The coordinate of C is
(a) (4, 3) (b) (4, 15) (c) (–4, –15) (d) (–15, –4)

5. The coordinates of the centroid of ΔABC with vertices A(–1, 0), B(5, –2) and C(8, 2) is
(a) (12, 0) (b) (6, 0) (c) (0, 6) (d) (4, 0)

6. If the points A(2, 3), B(5, k) and C(6, 7) are collinear, then the value of k is
(a) 4 (b) 6 (c) $\frac{-3}{2}$ (d) $\frac{11}{4}$

7. If P(–1, 1) is the middle point of the line segment joining A(–3, b) and B(1, b + 4) then the value of b is
(a) 1 (b) –1 (c) 2 (d) 0

8. y–axis divides the join of P(–4, 2) and Q(8, 3) in the ratio
(a) 3 : 1 (b) 1 : 3 (c) 2 : 1 (d) 1 : 2

9. x–axis divides the join of A(2, –3) and B(5, 6) in the ratio
(a) 3 : 5 (b) 2 : 3 (c) 2 : 1 (d) 1 : 2

10. The point P(1, 2) divides the join of A(–2, 1) and B(7, 4) are in the ratio of
(a) 3 : 2 (b) 2 : 3 (c) 2 : 1 (d) 1 : 2

11. A point P divides the join of A(5, –2) and B(9, 6) are in the ratio 3 : 1. The coordinates of P are
(a) (4, 7) (b) (8, 4) (c) $(\frac{11}{2}, 5)$ (d) (12, 8)

12. What point on x – axis is equidistant from the points A(7, 6) and B(–3, 4)?
(a) (0, 4) (b) (–4, 0) (c) (3, 0) (d) (0, 3)

13. The distance of the point P(4, –3) from the origin is
(a) 1 unit (b) 7 units (c) 5 units (d) 3 units

14. The distance between the points A(2, –3) and B(2, 2) is
(a) 2 units (b) 4 units (c) 5 units (d) 3 units

15. Find the area of the triangle whose vertices are A(1, 2), B(–2, 3) and C(–3, –4)
(a) 11sq. units (b) 22 sq. units (c) 7 sq. units (d) 6.5 sq. units

MCQ WORKSHEET-II
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1. Find the area of the triangle whose vertices are A(2, 4), B(–3, 7) and C(–4, 5)
(a) 11sq. units (b) 22 sq. units (c) 7 sq. units (d) 6.5 sq. units

2. Find the area of the triangle whose vertices are A(10, –6), B(2, 5) and C(–1, 3)
(a) 12.5 sq. units (b) 24.5 sq. units (c) 7 sq. units (d) 6.5 sq. units

3. Find the area of the triangle whose vertices are A(4, 4), B(3, –16) and C(3, –2)
(a) 12.5 sq. units (b) 24.5 sq. units (c) 7 sq. units (d) 6.5 sq. units

4. For what value of x are the points A(–3, 12), B(7, 6) and C(x, 9) collinear?
(a) 1 (b) –1 (c) 2 (d) –2

5. For what value of y are the points A(1, 4), B(3, y) and C(–3, 16) collinear?
(a) 1 (b) –1 (c) 2 (d) –2

6. Find the value of p for which the points A(–1, 3), B(2, p) and C(5, –1) collinear?
(a) 1 (b) –1 (c) 2 (d) –2

7. What is the midpoint of a line with endpoints (–3, 4) and (10, –5)?
(a) (–13, –9) (b) (–6.5, –4.5) (c) (3.5, –0.5) (d) none of these

8. A straight line is drawn joining the points (3, 4) and (5,6). If the line is extended, the ordinate of the point on the line, whose abscissa is –1 is
(a) 1 (b) –1 (c) 2 (d) 0

9. If the distance between the points (8, p) and (4, 3) is 5 then value of p is
(a) 6 (b) 0 (c) both (a) and (b) (d) none of these

10. The fourth vertex of the rectangle whose three vertices taken in order are (4,1), (7, 4), (13, –2) is
(a) (10, –5) (b) (10, 5) (c) (8, 3) (d) (8, –3)

11. If four vertices of a parallelogram taken in order are (–3, –1), (a, b), (3, 3) and (4, 3). Then a : b =
(a) 1 : 4 (b) 4 : 1 (c) 1 : 2 (d) 2 : 1

12. Area of the triangle formed by (1, – 4), (3, – 2) and (– 3,16) is
(a) 40 sq. units (b) 48 sq. units (c) 24 sq. units (d) none of these

13. The points (2, 5), (4, - 1), (6, - 7) are vertices of an _____ triangle
(a) isosceles (b) equilateral (c) scalene (d) right angled

14. The area of triangle formed by the points (p, 2 - 2p), (l-p,2p) and (-4-p, 6- 2p) is 70 sq. units. How many integral value of p are possible ?
(a) 2 (b) 3 (c) 4 (d) none of these

15. If the origin is the mid-point of the line segment joined by the points (2,3) and (x,y), then the value of (x,y) is
(a) (2, –3) (b) (2, 3) (c) (–2, 3) (d) (–2, –3)

MCQ WORKSHEET-III
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1. The distance of the point P(2, 3) from the x-axis is:
(a) 2 (b) 3 (c) 1 (d) 5

2. The distance between the points A(0, 6) and B(0, -2) is:
(a) 2 (b) 6 (c) 4 (d) 8

3. The distance of the point P(-6, 8) from the origin is:
(a) 8 (b) 27 (c) 10 (d) 6

4. The distance between the points (0, 5) and (-5, 0) is:
(a) 5 (b) 52 (c) 25 (d) 10

5. AOBC is a rectangle whose three vertices are A(0, 3), O(0, 0) and B(5, 0). The length of its diagonal is:
(a) 5 (b) 3 (c) 34 (d) 4

6. The perimeter of a triangle with vertices (0, 4), (0, 0) and (3, 0) is:
(a) 5 (b) 12 (c) 11 (d) 7 + 5

7. The area of a triangle with vertices A(3, 0), B(7, 0) and C(8, 4) is:
(a) 14 (b) 28 (c) 8 (d) 6

8. The points (–4, 0), (4, 0), (0, 3) are the vertices of a :
(a) Right triangle (b) Isosceles triangle (c) Equilateral triangle (d) Scalene triangle

9. Point on x – axis has coordinates:
(a) (a, 0) (b) (0, a) (c) (–a, a) (d) (a, –a)

10. Point on y – axis has coordinates:
(a) (–a, b) (b) (a, 0) (c) (0, b) (d) (–a, –b)

11. Line formed by joining (- 1,1) and (5, 7) is divided by a line x + y = 4 in the ratio of
(a) 1 : 4 (b) 1 : 3 (c) 1 : 2 (d) 3 : 4

12. If the area of the triangle with vertices (x, 0), (1,1) and (0, 2) is 4 square units, then a value of x is
(a) –2 (b) –4 (c) –6 (d) 8

MCQ WORKSHEET-IV
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1. Point A(–5, 6) is at a distance of:
(a) 61 units from the origin (b) 11 units from the origin
(c) $\sqrt{61}$ units from the origin (d) $\sqrt{11}$ units from the origin

2. If the points (1, x), (5, 2) and (9, 5) are collinear then the value of x is
(a) $\frac{5}{2}$ (b) $\frac{-5}{2}$ (c) –1 (d) 1

3. The end points of diameter of circle are (2, 4) and (–3, –1). The radius of the circle us
(a) $\frac{5\sqrt{2}}{2}$ (b) $5\sqrt{2}$ (c) $3\sqrt{2}$ (d) $\frac{\pm 5\sqrt{2}}{2}$

4. The ratio in which x – axis divides the line segment joining the points (5, 4) and (2, –3) is:
(a) 5 : 2 (b) 3 : 4 (c) 2 : 5 (d) 4 : 3

5. The point which divides the line segment joining the points (7, –6) and (3, 4) in ratio 1:2 internally lies in the
(a) I quadrant (b) II quadrant (c) III quadrant (d) IV quadrant

6. The point which lies on the perpendicular bisector of the line segment joining the points A(–2, –5) and B(2, 5) is:
(a) (0, 0) (b) (0, 2) (c) (2, 0) (d) (–2, 0)

7. The fourth vertex D of a parallelogram ABCD whose three vertices are A(-2, 3), B(6, 7) and C(8, 3) is:
(a) (0, 1) (b) (0, -1) (c) (-1, 0) (d) (1, 0)

8. If the point P(2, 1) lies on the line segment joining points A(4, 2) and B(8, 4), then
(a) AP = $\frac{1}{3}$AB (b) AP = PB (c) PB = $\frac{1}{3}$AB (d) AP = $\frac{1}{2}$AB

9. Three vertices of a parallelogram taken in order are (- 1, - 6), (2, - 5) and (7, 2). The fourth vertex is
(a) (1, 4) (b) (1, 1) (c) (4, 4) (d) (4, 1)

10. If A and B are the points (- 3, 4) and (2,1) respectively, then the coordinates of the points on AB produced such that AC = 2BC are
(a) (2, 4) (b) (3, 7) (c) (7, –2) (d) none of these

11. Distance of the point (4, a) from x-axis is half its distance from y-axis then a =
(a) 2 (b) 8 (c) 4 (d) 6

12. A triangle is formed by the points 0(0, 0), A(5,0) and B(0,5). The number of points having integral coordinates (both x and y) and strictly inside the triangle is
(a) 10 (b) 17 (c) 16 (d) 6

13. If P(l, 2), Q(4,6), R(5,7) and S(a, b) are the vertices of a parallelogram PQRS then
(a) a = 2, b = 4 (b) a = 3, b = 4 (c) a = 2, b = 3 (d) a = 3, b = 5

14. The number of points on x-axis which are at a distance of 2 units from (2, 4) is
(a) 2 (b) 1 (c) 3 (d) 0

15. The distance of the point (h, k) from x-axis is
(a) h (b) k (c) | h | (d) | k |

16. The vertices of a triangle are (0, 0), (3, 0) and (0, 4). Its orthocentre is at
(a) (0, 3) (b) (4, 0) (c) (0, 0) (d) (3, 4)

17. The area of the triangle with vertices at the points (a, b + c), (b, c + a) and (c, a + b) is
(a) a + b + c (b) a + b – c (c) a – b + c (d) 0

18. If the segment joining the points (a, b) and (c, d) subtends a right angle at the origin, then
(a) ac – bd = 0 (b) ac + bd = 0 (c) ab – cd = 0 (d) ab + cd = 0

19. The distance of A(5, -12) from the origin is
(a) 12 (b) 11 (c) 13 (d) 10

20. Find the ordinate of a point whose abscissa is 10 and which is at a distance of 10 units from the point P(2, -3).
(a) 3 (b) –9 (c) both (a) or (b) (d) none of these

PRACTICE QUESTIONS
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COORDINATE GEOMETRY
DISTANCE FORMULA

1. Find the distance between the following points:
(i) A(9, 3) and (15, 11) (ii) A(7, – 4) and b(–5, 1).
(iii) A(–6, –4) and B(9, –12) (iv) A(1, –3) and B(4, –6)
(v) P(a + b, a – b) and Q(a – b, a + b)
(vi) P(a sinα, a cosα) and Q(a cosα, –asinα)

2. If A(6, –1), B(1, 3) and C(k, 8) are three points such that AB= BC, find the value of k.

3. Find all the possible value of a for which the distance between the points A(a, –1) and B(5, 3) is 5 units.

4. Determine, whether each of the given points (–2, 1), (2, –2) and (5, 2) are the vertices of right angle.

5. Determine if the points (1, 5), (2, 3) and (– 2, – 11) are collinear.

6. By distance formula, show that the points (1, –1), (5, 2) and (9, 5) are collinear.

7. Find the value of *k* if the points A(2, 3), B(4, *k*) and C(6, –3) are collinear.

8. Find a relation between *x* and *y* if the points (*x*, *y*), (1, 2) and (7, 0) are collinear.

9. Find the point on x-axis which is equidistant from (–2, 5) and (2, –3).

10. Find the point on x-axis which is equidistant from (7, 6) and (–3, 4).

11. Find the point on the *x*-axis which is equidistant from (2, –5) and (–2, 9).

12. Find a point on the *y*-axis which is equidistant from the points A(6, 5) and B(– 4, 3).

13. Find a point on the *y*-axis which is equidistant from the points A(5, 2) and B(– 4, 3).

14. Find a point on the *y*-axis which is equidistant from the points A(5, – 2) and B(– 3, 2).

15. Find the point on y-axis, each of which is at a distance of 13 units from the point (–5, 7).

16. Find the point on x-axis, each of which is at a distance of 10 units from the point (11, –8).

17. Find the values of k for which the distance between the points A(k, –5) and B(2, 7) is 13 units.

18. Prove that the points A(–3, 0), B(1, –3) and C(4, 1) are the vertices of an isosceles right-angled triangle. Find the area of this triangle.

19. Prove that the points A(a, a), B(–a, –a) and C($-\sqrt{3}$ a, $\sqrt{3}$ a) are the vertices of an equilateral triangle. Calculate the area of this triangle.

20. If the distance of P(x, y) from A(5, 1) and B(–1, 5) are equal. Prove that 3x = 2y.

21. Show that the points A(1, 2), B(5, 4), C(3, 8) and D(–1, 6) are vertices of a square.

22. Show that the points A(5, 6), B(1, 5), C(2, 1) and D(6, 2) are vertices of a square.

23. Show that the points A(0, –2), B(3, 1), C(0, 4) and D(–3, 1) are vertices of a square. Also find its area.

24. Show that the points A(6, 2), B(2, 1), C(1, 5) and D(5, 6) are vertices of a square. Also find its area.

25. Show that the points A(–4, –1), B(–2, –4), C(4, 0) and D(2, 3) are vertices of a rectangle. Also find its area.

26. Prove that the points A(2, –2), B(14, 10), C(11, 13) and D(–1, 1) are vertices of a rectangle. Find the area of this rectangle.

27. Show that the points A(1, –3), B(13, 9), C(10, 12) and D(–2, 0) are vertices of a rectangle.

28. Show that the points A(1, 0), B(5, 3), C(2, 7) and D(–2, 4) are vertices of a rhombus.

29. Prove that the points A(2, –1), B(3, 4), C(–2, 3) and D(–3, –2) are vertices of a rhombus. Find the area of this rhombus.

30. Show that the points A(–3, 2), B(–5, –5), C(2, –3) and D(4, 4) are vertices of a rhombus. Find the area of this rhombus.

31. Prove that the points A(–2, –1), B(1, 0), C(4, 3) and D(1, 2) are vertices of a parallelogram.

32. Find the area of a rhombus if its vertices are (3, 0), (4, 5), (– 1, 4) and (– 2, – 1) taken in order.

33. Find the coordinates of the circumcentre of a triangle whose vertices are A(4, 6), B(0, 4) and C(6, 2). Also, find its circumradius.

34. Find the coordinates of the circumcentre of a triangle whose vertices are A(3, 0), B(–1, –6) and C(4, –1). Also, find its circumradius.

35. Find the coordinates of the circumcentre of a triangle whose vertices are A(8, 6), B(8, –2) and C(2, –2). Also, find its circumradius.

36. Find the coordinates of the centre of a circle passing through the points A(2, 1), B(5, –8) and C(2, –9). Also find the radius of this circle.

37. Find the coordinates of the centre of a circle passing through the points A(–2, –3), B(–1, 0) and C(7, –6). Also find the radius of this circle.

38. Find the coordinates of the centre of a circle passing through the points A(1, 2), B(3, –4) and C(5, –6). Also find the radius of this circle.

39. Find the coordinates of the centre of a circle passing through the points A(0, 0), B(–2, 1) and C(– 3, 2). Also find the radius of this circle.

40. Find the centre of a circle passing through the points (6, – 6), (3, – 7) and (3, 3).

41. Find the coordinates of the point equidistant from three given points A(5, 3), B(5, –5) and C(1, – 5).

42. If the points A(6, 1), B(8, 2), C(9, 4) and D(p, 3) are the vertices of a parallelogram, taken in order, find the value of p.

43. Find a relation between x and y such that the point (x , y) is equidistant from the points (7, 1) and (3, 5).

44. Check whether (5, – 2), (6, 4) and (7, – 2) are the vertices of an isosceles triangle.

45. Find the values of y for which the distance between the points P(2, – 3) and Q(10, y) is 10 units.

46. If Q(0, 1) is equidistant from P(5, –3) and R(x, 6), find the values of x. Also find the distances QR and PR.

47. If two vertices of an equilateral triangle be O(0, 0) and A(3, $\sqrt{3}$), find the coordinates of its third vertex.

48. The two opposite vertices of a square are (–1, 2) and (3, –2). Find the coordinates of the other two vertices.

49. The two opposite vertices of a square are (1, –6) and (5, 4). Find the coordinates of the other two vertices.

50. Prove that the points A(7, 10), B(–2, 5) and C(3, –4) are the vertices of an isosceles right triangle.

51. Show that the points A(–5, 6), B(3, 0) and C(9, 8) are the vertices of an isosceles right angled triangle. Find the area of this triangle.

52. Show that the points A(2, 1), B(5, 2), C(6, 4) and D(3, 3) are the angular points of parallelogram. Is this figure a rectangle?

53. Show that the points O(0, 0), A(3, $\sqrt{3}$) and B(3, $-\sqrt{3}$) are the vertices of an equilateral triangle. Find the area of this triangle.

54. Prove that the points A(3, 0), B(6, 4) and C(–1, 3) are the vertices of a right triangle. Also prove that these are vertices of an isosceles triangle.

55. If P and Q are two points whose coordinates are (at^2, 2at) and $\left(\frac{a}{t^2}, \frac{2a}{t}\right)$ respectively and S is the point (a, 0). Show that $\frac{1}{SP}+\frac{1}{SQ}$ is independent of t.

56. If the points A(4, 3) and B(x, 5) are on the circle with centre O(2, 3), find the value of x.

57. Find the relation between x and y if point P(x, y) lies on the perpendicular bisector of the line joining the points (3, 6) and (–3, 4).

58. Find the relation between x and y if point P(x, y) lies on the perpendicular bisector of the line joining the points (7, 1) and (3, 5).

59. If A, B and P are the points (–4, 3), (0, –2) and (α, β) respectively and P is equidistant from A and B, show that $8\alpha - 10\beta + 21 = 0$.

60. If the points (5, 4) and (x, y) are equidistant from the point (4, 5), prove that $x^2 + y^2 – 8x – 10y + 39 = 0$.

61. Find the coordinates of the point whish is at a distance of 2 units from (5, 4) and 10 units from (11, –2).

62. If two vertices of an equilateral triangle are (0, 0) and (3, 0), find the third vertex.

63. The centre of a circle is (2α–1, 3α+1) and it passes through the point (–3, –1). If a diameter of the circle is of length 20 units, find the value of α.

64. If the point P(x, y) is equidistant from the points A(5, 1) and B(–1, 5), prove that x = y.

65. Find the value of k if the point P(0, 2) is equidistant from (3, k) and (k, 5).

66. Let the opposite angular points of a square be (3, 4) and (1, –1). Find the coordinates of the remaining angular points.

67. Prove that the points (2x, 4a), (2a, 6a) and ($2a + \sqrt{3}a$, 5a) are the vertices of an equilateral triangle.

68. An equilateral triangle has two vertices at the points (3, 4) and (–2, 3), find the coordinates of the third vertex.

69. Two vertex of an isosceles triangle are (2, 0) and (2, 5). Find the third vertex if the length of the equal sides is 3.

70. The coordinates of the point P are (–3, 2). Find the coordinates of the point Q which lies on the line joining P and origin such that OP = OQ.

PRACTICE QUESTIONS
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SECTION FORMULA

1. Find the coordinates of the point which divides the line segment joining the points A(4, –3) and B(9, 7) in the ration 3 : 2.

2. Find the coordinates of the point which divides the line segment joining the points A(–1, 7) and B(4, –3) in the ration 2 : 3.

3. Find the coordinates of the point which divides the line segment joining the points A(–5, 11) and B(4, –7) in the ration 7 : 2.

4. Find the coordinates of the midpoint of the line segment joining the points A(–5, 4) and B(7, –8)

5. Find the coordinates of the midpoint of the line segment joining the points A(3, 0) and B(5, 4)

6. Find the coordinates of the midpoint of the line segment joining the points A(–11, –8) and B(8, –2).

7. The coordinates of the midpoint of the line segment joining the points A(2p + 1, 4) and B(5, q–1) are(2p, q). Find the value of p and q.

8. The midpoint of the line segment joining A(2a, 4) and B(–2, 3b) is M(1, 2a + 1). Find the values of a and b.

9. Find the coordinates of the points which divide the line segment joining the points (–2, 0) and (0, 8) in four equal parts.

10. Find the coordinates of the points which divide the line segment joining the points (–2, 2) and (2, 8) in four equal parts.

11. In what ratio does the points P(2,–5) divide the line segment joining A(–3, 5) and B(4, –9).

12. In what ratio does the points P(2, 5) divide the line segment joining A(8, 2) and B(–6, 9).

13. Find the coordinates of the points of trisection of the line segment joining the points (4, –1) and (–2, –3).

14. The line segment joining the points (3, –4) and (1, 2) is trisected at the points P(p, –2) and Q($\frac{5}{3}$, q). Find the values of p and q.

15. The coordinate of the midpoint of the line joining the point (3p, 4) and (–2, 2q) are (5, p). Find the value of p and q.

16. The consecutive vertices of a parallelogram ABCD are A(1, 2), B(1, 0) and C(4, 0). Find the fourth vertex D.

17. Find the lengths of the median of the triangle whose vertices are (1, –1), (0, 4) and (–5, 3).

18. Prove that the diagonal of a rectangle bisects each other and are equal.

19. Find the ratio in which the point (11, 15) divides the line segment joining the point (15, 5) and (9, 20).

20. Find the ratio in which the point P(m, 6) divides the line segment joining the point A(–4, 3) and B(2, 8). Also find the value of m.

21. If two vertices of ΔABC are A(3, 2), B(–2, 1) and its centroid G has the coordinate (5/3, –1/3). Find the coordinates of the third vertex.

22. The coordinate of the midpoint of the line joining the point (2p, 4) and (–2, 2q) are (3, p). Find the value of p and q.

23. Show that the points A(3, 1), B(0, –2), C(1, 1) and D(4, 4) are the vertices of a parallelogram ABCD.

24. If the points P(a, –11), Q(5, b), R(2, 15) and S(1, 1) are the vertices of a parallelogram PQRS, find the value of a and b.

25. If three consecutive vertices of a parallelogram ABCD are A(1, –2), B(3, 6) and C(5, 10). Find the fourth vertex D.

26. In what ratio does the point (– 4, 6) divide the line segment joining the points A(– 6, 10) and B(3, – 8)?

27. Find the coordinates of the points of trisection of the line segment joining the points A(2, – 2) and B(– 7, 4).

28. Find the ratio in which the *y*-axis divides the line segment joining the points (5, – 6) and (–1, – 4). Also find the point of intersection.

29. If the points A(6, 1), B(8, 2), C(9, 4) and D(*p*, 3) are the vertices of a parallelogram, taken in order, find the value of *p*.

30. If the points A(–2, –1), B(a, 0), C(4, b) and D(1, 2) are the vertices of a parallelogram, taken in order, find the value of *a and b*.

31. Find the ratio in which the point P(–6, a) divides the join of A(–3, –1) and B(–8, 9). Also, find the value of a.

32. Find the ratio in which the point P(–3, a) divides the join of A(–5, –4) and B(–2, 3). Also, find the value of a.

33. Find the ratio in which the point P(a, 1) divides the join of A(–4, 4) and B(6, –1). Also, find the value of a.

34. Find the ratio in which the line segment joining the points (– 3, 10) and (6, – 8) is divided by (– 1, 6).

35. Find the ratio in which the line segment joining A(1, – 5) and B(– 4, 5) is divided by the *x*-axis. Also find the coordinates of the point of division.

36. In what ratio is the line segment joining A(6, –3) and B(–2, –5) is divided by the *x*-axis. Also find the coordinates of the point of intersection of AB and the x-axis.

37. In what ratio is the line segment joining A(2, –3) and B(5, 6) is divided by the *x*-axis. Also find the coordinates of the point of intersection of AB and the x-axis.

38. In what ratio is the line segment joining A(–2, –3) and B(3, 7) is divided by the *y*-axis. Also find the coordinates of the point of intersection of AB and the y-axis.

39. The coordinates of one end point of a diameter AB of a circle are A(4, –1) and the coordinates of the centre of the circle are C(1, –3).

40. Find the coordinates of a point A, where AB is the diameter of a circle whose centre is (2, – 3) and B is (1, 4).

41. The line segment joining A(–2, 9) and B(6, 3) is a diameter of a circle with centre C. Find the coordinates of C.

42. AB is a diameter of a circle with centre C(–1, 6). If the coordinates of A are (–7, 3), find the coordinates of B.

43. Find the ratio in which the line $2x + y - 4 = 0$ divides the line segment joining the points A(2, –2) and B(3, 7).

44. Find the ratio in which the line $x - y - 2 = 0$ divides the line segment joining the points A(3, –1) and B(8, 9).

45. Find the ratio in which the line $3x + 4y - 9 = 0$ divides the line segment joining the points A(1, 3) and B(2, 7).

46. Find the lengths of the medians of a triangle ABC whose vertices are A(7, –3), B(5, 3) and C(3, –1).

47. Find the lengths of the medians of a triangle ABC whose vertices are A(0, –1), B(2, 1) and C(0,3).

48. Let D(3, –2), E(–3, 1) and F(4, –3) be the midpoints of the sides BC, CA and AB respectively of ΔABC. Then, find the coordinates if the vertices A, B and C.

49. If A and B are (– 2, – 2) and (2, – 4), respectively, find the coordinates of P such that AP = $\frac{3}{7}$ AB and P lies on the line segment AB.

50. A(1, 1) and B(2, –3) are two points. If C is a point lying on the line segment AB such that CB = 2AC, find the coordinates of C

51. If A(1, 1) and B(–2, 3) are two points and C is a point on AB produced such that AC = 3AB, find the coordinates of C.

52. Find the coordinates of the point P which is three-fourth of the way from A(3, 1) to B(–2, 5).

53. The line joining the points A(4, –5) and B(4, 5) is divided by the point P such that AP : AB = 2 : 5, find the coordinates of P.

54. The point P(–4, 1) divides the line segment joining the points A(2, –2) and B in the raio 3 : 5. Find the point B.

55. If A and B are (4, –5) and (4, 5), respectively, find the coordinates of P such that AP = $\frac{2}{5}$ AB and P lies on the line segment AB.

56. Find the coordinates of the points of trisection of the line segment AB, whose end points are A(2, 1) and B(5, –8).

57. Find the coordinates of the points which divide the join A(–4, 0) and B(0, 6) in three equal parts.

58. The line joining the points A(2, 1) and B(5, –8) is trisected at the points P and Q. If the point P lies on the line $2x – y + k = 0$, find the value of k.

59. Find the coordinates of the points which divide the line segment joining A(– 2, 2) and B(2, 8) into four equal parts.

60. If A(5, –1), B(–3, –2) and C(–1, 8) are the vertices of ΔABC, find the length of the median through A and the coordinates of the centroid.

61. Find the centroid of ΔABC whose vertices are vertices are A(–3, 0), B(5, –2) and C(–8, 5).

62. Two vertices of a ΔABC are given by A(6, 4) and B(–2, 2) and its centroid is G(3, 4). Find the coordinates of the third vertex C of ΔABC.

63. Find the coordinates of the centroid of a ΔABC whose vertices are A(6, –2), B(4, –3) and C(–1, –4).

64. Find the centroid of a ΔABC whose vertices are A(–1, 0), B(5, –2) and C(8, 2).

65. A(3, 2) and B(–2, 1) are two vertices of a ΔABC, whose centroid is G($\frac{5}{3}, \frac{-1}{3}$). Find the coordinates of the third vertex C.

66. If G(–2, 1) is the centroid of ΔABC and two of its vertices are A(1, –6) and B(–5, 2), find the third vertex of the triangle.

67. Find the third vertex of a ΔABC if two of its vertices are B(–3, 1) and C(0, –2) and its centroid is at the origin.

68. The line segment joining A(–1, $\frac{5}{3}$) and B(a, 5) is divided in the ratio 1 : 3 at P, the point where the line segment AB intersects y-axis. Find the value of a and the coordinates of P.

69. Find the ratio in which the point P whose ordinate is –3 divides the join of A(–2, 3) and B(5, $\frac{-15}{2}$). Hence find the coordinate of P.

70. Calculate the ratio in which the line joining the points A(6, 5) and B(4, –3) is divided by the line $y = 2$. Also, find the coordinates of the point of intersection.

71. Show that the points (3, –2), (5, 2) and (8, 8) are collinear by using section formula.

72. If the points (–1, –1), (2, p) and (8, 11) are collinear, find the value of p using section formula.

73. If the points (2, 3), (4, k) and (6, –3) are collinear, find the value of k using section formula.

74. If two vertices of a parallelogram are (3, 2), (–1, 0) and its diagonals meet at (2, –5), find the other two vertices of the parallelogram.

75. Find the coordinates of the vertices of a triangle whose midpoints are (–3, 2), (1, –2) and (5, 6).

76. Find the third vertex of a triangle if its two vertices are (–1, 4) and (5, 2) and midpoint of one side is (0, 3).

77. If the midpoints of the sides of a triangle are (2, 3), $(\frac{3}{2}, 4)$ and $(\frac{11}{2}, 5)$, find the centroid of the triangle.

78. If the points (10, 5), (8, 4) and (6, 6) are the midpoints of the sides of a triangle, find its vertices.

79. If the point C(–1, 2) divides the line segment AB in the ratio 3 : 4, where the coordinates of A are (2, 5), find the coordinates of B.

80. The vertices of a quadrilateral are (1, 4), (–2, 1), (0, –1) and (3, 2). Show that diagonals bisect each other. What does quadrilateral become?

81. Using analytical geometry, prove that the midpoint of the hypotenuse of a right triangle is equidistant from its vertices.

82. Using analytical geometry, prove that the diagonals of a rhombus are perpendicular to each other.

83. Prove analytically that the line segment joining the midpoint of two sides of a triangle is half of the third side.

84. If (–2, 3), (4, –3) and (4, 5) are the midpoints of the sides of a triangle, find the coordinates of its centroid.

85. If (1, 1), (2, –3) and (3, 4) are the midpoints of the sides of a triangle, find the coordinates of its centroid.

PRACTICE QUESTIONS
CLASS X : CHAPTER – 7
COORDINATE GEOMETRY
AREA OF TRIANGLE

1. Find the area of a triangle formed by the points A(5, 2), B(4, 7) and C(7, – 4).

2. Find the area of a triangle formed by the points A(1, –1), B(– 4, 6) and C(– 3, – 5).

3. Find the area of a triangle formed by the points A(2, 3), B(– 1, 0) and C(2, –4).

4. Find the area of a triangle formed by the points A(10, –6), B(2, 5) and C(– 1, 3).

5. Determine if the points (1, 5), (2, 3) and (– 2, – 11) are collinear.

6. Show that the points $(\frac{-3}{2}, 3)$, (6, –2), (–3, 4) are collinear by using area of triangle.

7. By using area of triangle show that the points (a, b + c), (b, c + a) and (c, a + b) are collinear.

8. Find the value of *k* if the points A(8, 1), B(k, –4) and C(2, –5) are collinear.

9. Find the value of *k* if the points A(7, –2), B(5, 1) and C(3, k) are collinear.

10. If A(3, 2), B(–1, 0) and C(1, –12) are the vertices of a triangle and D is midpoint of BC, find the coordinates of the point D. Also find the areas of ΔABD and ΔACD. Hence verify that the median AD divides the triangle ABC into two triangles of equal areas.

11. Find the value of *k* if the points A(2, 3), B(4, *k*) and C(6, –3) are collinear.

12. If A(–5, 7), B(– 4, –5), C(–1, –6) and D(4, 5) are the vertices of a quadrilateral, find the area of the quadrilateral ABCD.

13. If A(2, 1), B(6, 0), C(5, –2) and D(–3, –1) are the vertices of a quadrilateral, find the area of the quadrilateral ABCD.

14. If A(–4, 5), B(0, 7), C(5, –5) and D(–4, –2) are the vertices of a quadrilateral, find the area of the quadrilateral ABCD.

15. If A(0, 0), B(6, 0), C(4, 3) and D(0, 3) are the vertices of a quadrilateral, find the area of the quadrilateral ABCD.

16. If A(–4, –2), B(–3, –5), C(3, –2) and D(2, 3) are the vertices of a quadrilateral, find the area of the quadrilateral ABCD.

17. Find the area of the triangle formed by joining the mid-points of the sides of the triangle whose vertices are (0, –1), (2, 1) and (0, 3). Find the ratio of this area to the area of the given triangle.

18. Find a relation between *x* and *y* if the points (*x*, *y*), (1, 2) and (7, 0) are collinear.

19. If A(2, 1), B(–2, 3) and C(4, –3) are the vertices of a ΔABC and D, E are the midpoints of the sides AB, AC respectively, find the coordinates of D and E. Prove that the area of ΔABC is four times the area of ΔADE.

20. If A(4, 4), B(3, –16) and C(3, –2) are the vertices of a ΔABC and D, E, F are the midpoints of the sides BC, CA and AB respectively. Prove that the area of ΔABC is four times the area of ΔDEF.

21. Find the point P on the x – axis which is equidistant from the points A(5, 4) and B(–2, 3). Also find the area of ΔPAB.

22. If P(x, y) is any point on the line joining the points A(a, 0) and B(0, b), then show that $\frac{x}{a}+\frac{y}{b}=1$.

23. If the vertices of a triangle are (1, k), (4, –3) and (–9, 7) and its area is 15 sq. units, find the value(s) of k.

24. Find the value of m for which the points with coordinates (3, 5), (m, 6) and $\left(\frac{1}{2},\frac{15}{2}\right)$ are collinear.

25. Find the value of k for which the points with coordinates (2, 5), (4, 6) and $\left(k,\frac{11}{2}\right)$ are collinear.

26. Find the point P on x-axis which is equidistant from A(–2, 5) and B(2, –3). Also find the area of ΔPAB.

27. Find the point P on x-axis which is equidistant from A(7, 6) and B(–3, 4). Also find the area of ΔPAB.

28. Find the point P on the *x*-axis which is equidistant from A(2, –5) and B(–2, 9). Also find the area of ΔPAB.

29. Find a point P on the *y*-axis which is equidistant from the points A(6, 5) and B(– 4, 3). Also find the area of ΔPAB.

30. Find a point P on the *y*-axis which is equidistant from the points A(5, 2) and B(– 4, 3). Also find the area of ΔPAB.

31. Find a point P on the *y*-axis which is equidistant from the points A(5, – 2) and B(– 3, 2). Also find the area of ΔPAB.

32. Find the value of k for which the area formed by the triangle with vertices A(k, 2k), (–2, 6) and C(3, 1) is 5 square units.

33. Find the value of k for which the area formed by the triangle with vertices A(1, 2), (–2, 3) and C(–3, k) is 11 square units.

34. Find the value of k for which the area formed by the triangle with vertices A(4, 4), (3, k) and C(3, –2) is 7 square units.

35. For what value of p are the points A(–3, 9), B(2, p) and C(4, –5) are collinear.

36. Prove that the area of triangle whose vertices are (t, t – 2), (t + 2, t + 2) and (t + 3, t) is independent of t.

37. For what value of k are the points (k, 2 – 2k), (–k + 1, 2k) and (–4 – k, 6 – 2k) are collinear.

38. Find the condition that the point (x, y) may lie on the line joining (3, 4) and (–5, –6).

39. If the coordinates of two points A and B are (3, 4) and (5, –2) respectively. Find the coordinates of any point P, if PA = PB and area of ΔPAB = 10 sq. units.

40. The coordinates of A, B, C are (6, 3), (–3, 5) and (4, –2) respectively and P is any point (x, y). Show that the ratio of the areas of triangles PBC and ABC is $\left|\frac{x+y-2}{7}\right|$.

41. If (x, y) be on the line joining the two points (1, –3) and (–4, 2), prove that x + y + 2 = 0.

42. Prove that the points (a, b), (x, y) and (a – x, b – y) are collinear if ay = bx.

43. Four points A(6, 3), B(–3, 5), C(4, –2) and D(x, 3x) are given in such a way that $\frac{\Delta DBC}{\Delta ABC}=\frac{1}{2}$, find the value of x.

44. If three points (a, b), (c, d) and (e, f) are collinear, prove that $\frac{d-f}{ce}+\frac{f-b}{ea}+\frac{b-d}{ac}=0$.

45. The area of triangle is 5 sq. units. Two of its vertices are (2, 1) and (3, –2). The third vertex lies on y = x + 3. Find the third vertex.